Boundary-Layer Transfer and Bounded Sharp Saint–Venant (Route δ, Plan 2, Building Block BoundaryLayerTransfer)

Abstract

We close the route-δ chain *relative to the upstream metric trace input* by transferring the metric-trace estimate at the deficit endpoint $\widehat{\rho} := \rho_\delta = 1 - \kappa\sqrt{\delta_T}$ back to the original set Ω. The upstream weighted metric trace ([5]) supplies the uniform pointwise bound $\sup_{\rho\in J^*} d_{\mathcal{X}}(F_\rho, B_1)^2 \le C_*\,\delta_T(\Omega)$ on the order-one window $J^* = [\rho_\delta - L^*, \rho_\delta]$; specialising to the upper endpoint gives $d_{\mathcal{X}}(F_{\rho_\delta}, B_1)^2 \le C_*\,\delta_T(\Omega)$. The boundary-layer volume $|\Omega \setminus E_{\rho_\delta}| = \omega_n(1-\rho_\delta^n)$ is exactly $O(\sqrt{\delta_T})$ by the definition of ρ_δ; the elementary superlevel-transfer estimate of [1] (Lemma 8.3) gives $\mathcal{A}(\Omega) \le \mathcal{A}(E_{\rho_\delta}) + C_n\sqrt{\delta_T}$. Squaring with $(a+b)^2 \le 2a^2 + 2b^2$ doubles the constant on δ_T but preserves the sharp exponent 2. We obtain the bounded sharp Saint–Venant inequality $\mathcal{A}(\Omega)^2 \le C(n,R,\rho_*)\,\delta_T(\Omega)$ for all open $\Omega \subset B_R$ with $|\Omega| = \omega_n$ inside an explicit smallness window $\delta_T(\Omega) \le \delta_0(n,R,\rho_*)$. Outside that window the inequality is trivial after enlarging C via the FMP quantitative isoperimetric inequality applied to Ω. Every constant is polynomial in $1/\rho_*$, R, n and the constants of the upstream building blocks.

1 Statement of the theorem

Throughout, $n \ge 2$, $R \ge 2$, $\rho_* \in (0,1)$ are fixed. $\Omega \subset \mathbb{R}^n$ denotes a bounded open set with $|\Omega| = \omega_n$ and $\Omega \subset B_R$. The torsion functional is normalised by $\delta_T(\Omega) := E(\Omega) - E(B_1) \ge 0$. The Fraenkel asymmetry (cf. [14]) is

$$\mathcal{A}(\Omega) := \inf\left\{ \frac{|\Omega\Delta(z+B_1)|}{\omega_n} \ :\ z \in \mathbb{R}^n \right\} \in [0,2).$$

For $\rho \in [\rho_*, 1]$ we write $E_\rho = \{u_\Omega > t(\rho)\}$ where $|E_\rho| = \omega_n\rho^n$ and u_Ω is the torsion function of Ω (see [1]). The normalised level set $F_\rho := \rho^{-1}E_\rho \in \mathcal{X}$, where $\mathcal{X}$ is the metric quotient of finite-measure sets by translations equipped with

$$d_{\mathcal{X}}([A],[C]) := \inf_{a\in\mathbb{R}^n} |A\Delta(C+a)|.$$

Theorem 1.1 (Bounded sharp Saint–Venant; building block BoundaryLayerTransfer)**.** *There exist $C = C(n,R,\rho_*) > 0$ and $\delta_0 = \delta_0(n,R,\rho_*) > 0$ such that every open $\Omega \subset B_R$ with $|\Omega| = \omega_n$ satisfies*

$$\mathcal{A}(\Omega)^2 \le C(n,R,\rho_*)\,\delta_T(\Omega). \tag{1}$$

The constant $C(n,R,\rho_*)$ is polynomial in $1/\rho_*$, R, n and the constants of [5]. The exponent 2 is sharp. We further assume Ω is open of full measure $|\Omega| = \omega_n$ contained in B_R, sufficient that the torsion function $u_\Omega \in W_0^{1,2}(\Omega)$ exists with level sets E_ρ of finite perimeter for a.e. ρ ([11] coarea).

The proof occupies §2–§7: the small-deficit regime $\delta_T(\Omega) \le \delta_0$ is closed by the metric trace, the elementary superlevel transfer, and the squaring identity; the complementary regime $\delta_T(\Omega) \ge \delta_0$ is closed by the trivial bound $\mathcal{A}(\Omega) \le 2$ after absorbing $4/\delta_0$ into C.

2 The metric-trace input

We record, in the form needed below, the output of the weighted metric trace lemma assembled in [5]. Set

$$\rho_\delta := 1 - \kappa\sqrt{\delta_T(\Omega)}, \tag{2}$$

where $\kappa = \kappa(n, \rho_*) > 0$ is the profile-gap constant fixed in [1]; equivalently ρ_δ is the largest radius for which the profile-gap lower bound on the measure $d\mu(\rho) = (-t'(\rho))\, d\rho$ holds (cf. [1, (5.1)]).

Remark 2.1 (No pointwise lower-gradient input)**.** The constant $C_* = C_*(n, R, \rho_*)$ inherited from [5, Thm. 1.3] no longer depends on a pointwise lower bound for $|\nabla u|$ on level surfaces. The former Hyp-G input has been replaced upstream by the self-truncated velocity estimate of [6]: on Fusco–Julin good levels, the low-gradient part of $\partial^* E_\rho$ is charged by the Cauchy–Schwarz defect D_H itself. The boundary-layer transfer arguments of the present block do not use any gradient lower bound directly.

The admissible class of Ω here is the output of the bounded reduction [8], which produces $\Omega \subset B_{R_*(n)}$ with $|\Omega| = \omega_n$.

Theorem 2.2 (Metric trace at ρ_δ; upstream input from [5])**.** *There exist $C_* = C_*(n, R, \rho_*) > 0$ and $\delta_1 = \delta_1(n, R, \rho_*) > 0$ such that for every open $\Omega \subset B_R$ with $|\Omega| = \omega_n$ and $\delta_T(\Omega) \le \delta_1$, setting*

$$\widehat{\rho} \;:=\; \rho_\delta \;=\; 1 - \kappa\sqrt{\delta_T(\Omega)}, \tag{3}$$

we have

$$d_{\mathcal{X}}(F_{\widehat{\rho}}, B_1)^2 \le C_*\, \delta_T(\Omega). \tag{4}$$

In fact the bound holds uniformly for $\rho \in J^* = [\rho_\delta - L^*, \rho_\delta]$ with $L^* = (\rho_\delta - \rho_*)/4$.

Reference. This is the conclusion of [5, Thm. 5.6, Cor. 5.7], combining the integrated metric-derivative bound [5, (4.2)] (which encodes the integrated Cauchy–Schwarz defect D_H) with the profile-gap measure [1, (5.1)] and the Fusco–Julin isoperimetric input [5, Lemma 3.1] (which encodes the integrated isoperimetric defect D_I). The bound at the endpoint ρ_δ follows from Markov on the distance term plus kinetic Cauchy–Schwarz transport across J^*; see [5, Rem. 5.7] for the structural exposition. No additional input is needed here.

Remark 2.3. By the definition of $d_{\mathcal{X}}$ and the volume normalisation $|E_{\widehat{\rho}}| = \omega_n \widehat{\rho}^n$, there exists $z = z(\widehat{\rho}) \in \mathbb{R}^n$ such that $|E_{\widehat{\rho}} \Delta (z + B_{\widehat{\rho}})| \le \widehat{\rho}^n d_{\mathcal{X}}(F_{\widehat{\rho}}, B_1)$. Dividing by $|E_{\widehat{\rho}}| = \omega_n \widehat{\rho}^n$,

$$\mathcal{A}(E_{\widehat{\rho}}) \le \omega_n^{-1} d_{\mathcal{X}}(F_{\widehat{\rho}}, B_1), \qquad \mathcal{A}(E_{\widehat{\rho}})^2 \le \omega_n^{-2} C_* \delta_T(\Omega). \tag{5}$$

Here $\mathcal{A}(E_{\widehat{\rho}})$ is the Fraenkel asymmetry of $E_{\widehat{\rho}}$ relative to balls of *its own volume* $\omega_n \widehat{\rho}^n$, not of ω_n.

3 Volume of the removed boundary layer

Lemma 3.1 (Boundary-layer volume)**.** *Let $\widehat{\rho} = \rho_\delta$ as in Theorem 2.2. There is $C_n > 0$ depending only on n such that, for $\delta_T(\Omega) \le \delta_2(n, \rho_*)$ small,*

$$|\Omega \setminus E_{\widehat{\rho}}| \;=\; \omega_n\big(1 - \widehat{\rho}^n\big) \;\le\; C_n \sqrt{\delta_T(\Omega)}. \tag{6}$$

Proof. By (2) and (3), $\widehat{\rho} = 1 - \kappa\sqrt{\delta_T(\Omega)}$. The map $\rho \mapsto 1 - \rho^n$ is C^1 with derivative $-n\rho^{n-1}$, so the mean-value theorem applied on $[\widehat{\rho}, 1] \subset [\rho_*, 1]$ gives

$$1 - \widehat{\rho}^n \;\le\; n(1 - \widehat{\rho}) \;=\; n\kappa\sqrt{\delta_T(\Omega)}.$$

This proves (6) with $C_n := \omega_n n\kappa$. Equivalently, $C_n' := n\kappa$ so that $C_n = \omega_n C_n'$ and the prefactor in (15) simplifies to $2C_n'$ without double-cancellation of ω_n. Choosing $\delta_2 := \delta_1$ ensures (3) applies. □

Remark 3.2 (Why $\widehat{\rho} = \rho_\delta$ is the natural choice)**.** The upstream weighted metric trace [5, Thm. 5.6, Rem. 5.7] delivers a uniform pointwise bound $d_{\mathcal{X}}(F_\rho, B_1)^2 \le C_* \delta_T$ valid for *every* ρ in the order-one window $J^* = [\rho_\delta - L^*, \rho_\delta]$. The boundary-layer transfer cost is monotone in $\widehat{\rho}$: smaller $\widehat{\rho}$ leaves a larger boundary layer $|\Omega \setminus E_{\widehat{\rho}}| \asymp 1 - \widehat{\rho}$. Since both costs — the metric-trace bound and the layer-volume bound — balance optimally at $\widehat{\rho} = \rho_\delta$, where $1 - \widehat{\rho} = \kappa\sqrt{\delta_T}$ is the smallest layer in J^*, the endpoint choice is the unique sharp one.

4 The asymmetry transfer lemma

We now prove the elementary superlevel transfer inequality ([1], Lemma 8.3). The point is that the asymmetry of Ω at scale $|\Omega| = \omega_n$ is dominated by the asymmetry of any nested superlevel E_t at its own scale $|E_t| = \omega_n \rho^n$ plus a volume-defect correction; the ball-scale correction $|B_{\omega_n} \setminus B_s|$ is controlled by the same defect by elementary monotonicity.

Throughout this section, Ω has $|\Omega| = L = \omega_n$, $E_t \subset \Omega$ is open with $|E_t| = s$ and $\eta := L - s \in [0, L]$. We write B_r for the open ball of radius r centred at the origin and recall that the volume of B_r is $\omega_n r^n$, so the ball of volume s is B_{r_s} with $r_s := (s/\omega_n)^{1/n}$, and the ball of volume L is B_1.

Lemma 4.1 (Superlevel asymmetry transfer; cf. [1] Lemma 8.3)**.** *Let Ω, E_t be as above with $E_t \subset \Omega$ and $\eta = |\Omega| - |E_t|$. Then*

$$\mathcal{A}(\Omega) \le \mathcal{A}(E_t) + \frac{2\eta}{L}. \tag{7}$$

Equivalently, since $L = \omega_n$,

$$\mathcal{A}(\Omega) \le \mathcal{A}(E_t) + \frac{2}{\omega_n} |\Omega \setminus E_t|. \tag{8}$$

Proof. Pick $\varepsilon \in (0, L)$ and choose $z \in \mathbb{R}^n$ such that

$$|E_t \Delta (z + B_{r_s})| \le s\left(\mathcal{A}(E_t) + \varepsilon\right). \tag{9}$$

We compare Ω to the ball $z + B_1$ of volume L. The triangle inequality for symmetric differences gives

$$|\Omega \Delta (z + B_1)| \le |\Omega \Delta E_t| + |E_t \Delta (z + B_{r_s})| + |(z + B_{r_s}) \Delta (z + B_1)|. \tag{10}$$

We bound each summand.

First summand. Because $E_t \subset \Omega$, $\Omega \Delta E_t = \Omega \setminus E_t$, hence

$$|\Omega \Delta E_t| = \eta. \tag{11}$$

Second summand. (9) gives

$$|E_t \Delta (z + B_{r_s})| \le s\big(\mathcal{A}(E_t) + \varepsilon\big). \tag{12}$$

Third summand. Since $z + B_{r_s} \subset z + B_1$,

$$|(z + B_{r_s}) \Delta (z + B_1)| = |B_1 \setminus B_{r_s}| = L - s = \eta. \tag{13}$$

Combining (10)–(13) and dividing by L,

$$\frac{|\Omega \Delta (z + B_1)|}{L} \le \frac{2\eta}{L} + \frac{s}{L}\big(\mathcal{A}(E_t) + \varepsilon\big) \le \frac{2\eta}{L} + \mathcal{A}(E_t) + \varepsilon, \tag{14}$$

using $s \le L$. Since $\mathcal{A}(\Omega) \le L^{-1} |\Omega \Delta (z + B_1)|$ and $\varepsilon \in (0, L)$ is arbitrary, (7) follows. □

Remark 4.2 (On the ball-scale correction)**.** The triangle inequality (10) contains two distinct volume-defect terms: $|\Omega\Delta E_t| = \eta$ (the boundary layer of Ω outside E_t) and $|B_1 \setminus B_{r_s}| = \eta$ (the "ball-scale" correction). They add to give $2\eta/L$. There is no $|B_s|/|B_L|$ rescaling to worry about, because the symmetric difference $|E_t\Delta(z + B_{r_s})|$ is measured at the natural scale s of E_t, and the cross term to scale L is precisely $|B_1 \setminus B_{r_s}|$. This is the same combinatorial identity used in [1, Lemma 8.3] and in the L^1-rearrangement bookkeeping in [11] (see also [15]).

Corollary 4.3 (Transfer at the level $\widehat{\rho}$)**.** *Let $\Omega \subset B_R$ with $|\Omega| = \omega_n$, $\delta_T(\Omega) \le \delta_2$, and let $\widehat{\rho}$ be as in Theorem 2.2. Then*

$$\mathcal{A}(\Omega) \le \mathcal{A}(E_{\widehat{\rho}}) + 2C'_n\sqrt{\delta_T(\Omega)}, \tag{15}$$

where $C'_n = n(\kappa + C_*)$ (equivalently $C_n = \omega_n C'_n$) is the constant of Lemma 3.1.

Proof. $E_{\rho_\delta} \subset \Omega$ by definition, and $|\Omega \setminus E_{\rho_\delta}| \le C_n\sqrt{\delta_T(\Omega)}$ by Lemma 3.1. Apply (8) with $E_t = E_{\rho_\delta}$. ☐

5 Squaring preserves the exponent 2

We combine Corollary 4.3 with (5) and the elementary inequality $(a + b)^2 \le 2a^2 + 2b^2$.

Theorem 5.1 (Small-δ_T bound)**.** *There exist $C_3 = C_3(n, R, \rho_*) > 0$ and $\delta_0 = \delta_0(n, R, \rho_*) \in (0, \delta_2]$ such that every open $\Omega \subset B_R$ with $|\Omega| = \omega_n$ and $\delta_T(\Omega) \le \delta_0$ satisfies*

$$\mathcal{A}(\Omega)^2 \le C_3(n, R, \rho_*)\, \delta_T(\Omega). \tag{16}$$

Proof. Take $\delta_T(\Omega) \le \delta_2$ so that Corollary 4.3 applies. Set $a := \mathcal{A}(E_{\widehat{\rho}})$, $b := 2C'_n\sqrt{\delta_T(\Omega)}$. Then (15) reads $\mathcal{A}(\Omega) \le a + b$. Squaring,

$$\mathcal{A}(\Omega)^2 \le 2a^2 + 2b^2 = 2\mathcal{A}(E_{\widehat{\rho}})^2 + 8(C'_n)^2\, \delta_T(\Omega). \tag{17}$$

By (5), $\mathcal{A}(E_{\widehat{\rho}})^2 \le \omega_n^{-2}C_*\delta_T(\Omega)$. Substituting,

$$\mathcal{A}(\Omega)^2 \le \left(\frac{2C_*}{\omega_n^2} + 8(C'_n)^2\right)\delta_T(\Omega) =: C_3\, \delta_T(\Omega). \tag{18}$$

This proves (16) with $\delta_0 := \delta_2$. The exponent 2 is preserved because $(a+b)^2 \le 2a^2+2b^2$ *doubles the constant* but does not raise either summand to a higher power. ☐

Remark 5.2 (The factor of 2)**.** The identity $(a + b)^2 = a^2 + 2ab + b^2 \le 2(a^2 + b^2)$, valid by $2ab \le a^2 + b^2$, is sharp up to the factor of 2 (saturated when $a = b$). Replacing it by Young's inequality with a parameter $\theta \in (0, 1)$, $(a+b)^2 \le (1+\theta^{-1})a^2 + (1+\theta)b^2$, allows the prefactor on $\mathcal{A}(E_{\widehat{\rho}})^2$ to be made arbitrarily close to 1, at the cost of inflating the coefficient on δ_T. This does not affect the exponent, only the constant C_3. For the bounded sharp Saint–Venant inequality the cleanest choice is $\theta = 1$, giving (18).

6 Constants chain

The constant $C_3 = C_3(n, R, \rho_*)$ produced by Theorem 5.1 unwinds as

$$C_3 = \frac{2C_*}{\omega_n^2} + 8(C'_n)^2, \qquad C'_n = n\kappa, \tag{19}$$

where the inputs are:

- $C_* = C_*(n, R, \rho_*)$: the metric trace constant of Theorem 2.2, supplied by [5, Theorem on weighted metric trace]. In the repaired metric-trace block, C_* is polynomial in $(1/\rho_*, R, n)$ and in the constants of the Fusco–Julin and oriented radial-trace inputs.
- $C_n = \omega_n n\kappa$ from Lemma 3.1, where $\kappa = \kappa(n, \rho_*)$ is the profile-gap constant of [1]. The simpler form (no C_* contribution) reflects the endpoint choice $\widehat{\rho} = \rho_\delta$, see Remark 3.2.
- $\omega_n = |B_1|$ is purely dimensional.

In particular, C_3 is polynomial in $(1/\rho_*, R, n)$ and in the explicit constants of the repaired metric trace. Specialising to $R = R_*(n)$ (the bounded reduction radius) and $\rho_* = 1/2$ as in [5], C_3 depends only on n.

Remark 6.1 (Sharpness of the dependence)**.** The polynomial dependence on $1/\rho_*$ enters through (i) the profile-gap constant in [1, (5.1)]; (ii) the Cauchy–Schwarz bookkeeping in the metric derivative bound [5, (2.4),(4.2)]; and (iii) the Fusco–Julin/oriented-radial-trace input to the homothetic velocity defect [5]. None of these can be improved beyond polynomial without altering the upstream construction.

7 Smallness regime parametrisation and the global bound

Theorem 5.1 closes the small-δ_T regime relative to the upstream metric-trace input. For the complementary regime $\delta_T(\Omega) \geq \delta_0$, the trivial bound $\mathcal{A}(\Omega) < 2$ suffices.

Lemma 7.1 (Large-δ_T trivial bound)**.** *For every open $\Omega \subset B_R$ with $|\Omega| = \omega_n$ and $\delta_T(\Omega) \geq \delta_0(n, R, \rho_*)$,*

$$\mathcal{A}(\Omega)^2 \leq \frac{4}{\delta_0(n, R, \rho_*)}\, \delta_T(\Omega). \tag{20}$$

Proof. $\mathcal{A}(\Omega) < 2$ by definition of the Fraenkel asymmetry, so $\mathcal{A}(\Omega)^2 < 4$. By assumption, $1 \leq \delta_T(\Omega)/\delta_0$, hence $\mathcal{A}(\Omega)^2 \leq 4 \leq 4\delta_T(\Omega)/\delta_0$. □

Remark 7.2 (Alternative via FMP)**.** A second proof of (20) uses the FMP quantitative isoperimetric/Saint–Venant inequality [10, 11] applied to Ω itself: $\mathcal{A}(\Omega)^4 \leq C_{\mathrm{FMP}}\delta_T(\Omega)$, which on $\delta_T(\Omega) \geq \delta_0$ implies a quadratic bound with constant $C'_{\mathrm{FMP}}/\delta_0^{1/2}$. The bound via $\mathcal{A} < 2$ is sharper for our purposes.

Proof of Theorem 1.1. For $\delta_T(\Omega) \leq \delta_0$, (1) holds with $C(n, R, \rho_*) := C_3(n, R, \rho_*)$ by Theorem 5.1. For $\delta_T(\Omega) \geq \delta_0$, Lemma 7.1 gives the same bound with $C(n, R, \rho_*) := 4/\delta_0(n, R, \rho_*)$. Taking

$$C(n, R, \rho_*) := \max\left\{ C_3(n, R, \rho_*), \frac{4}{\delta_0(n, R, \rho_*)} \right\} \tag{21}$$

proves (1) on the whole admissible class.

Sharpness of the exponent 2 is the standard ellipsoidal saturation: e.g., [13, Introduction] shows $\delta_T(\Omega_\varepsilon) \sim \varepsilon^2$ for the ellipsoidal one-parameter family Ω_ε. Along this family of volume-preserving ellipsoidal perturbations Ω_ε of B_1, $\mathcal{A}(\Omega_\varepsilon) \sim \varepsilon$ and $\delta_T(\Omega_\varepsilon) \sim \varepsilon^2$ to second order, so any bound $\mathcal{A}(\Omega)^p \leq C\delta_T(\Omega)$ requires $p \geq 2$. □

Remark 7.3 (How this slots into GlobalAssembly)**.** Theorem 1.1 is precisely the input to the bounded reduction step of [8, Theorem 5.1], specialised to $R = R_*(n)$. Together with the Kohler–Jobin transfer of [9, Theorem 6.1], it yields the sharp quantitative Faber–Krahn inequality

$$\big(|\Omega|/|B_1|\big)^{2/n}\big(\lambda_1(\Omega) - \lambda_1(B^*)\big) \geq c_{\mathrm{FK}}(n)\, \mathcal{A}(\Omega)^2,$$

as assembled in [7, §6].

Remark 7.4 (Where this block sits in the Plan 2 chain)**.** The boundary-layer transfer is the last *geometric* step in the route-δ chain. Upstream, [1] provides the exact deficit identity, [2] the metric first variation, and [5] the metric trace at $\widehat{\rho}$. The present block does *not* use any pointwise (R)-type input, only the metric trace (4) and the elementary triangle inequality of §4. In particular, no $C^{1,\gamma}$ regularity of $\partial\Omega$ is required; the proof is purely finite-perimeter.

References

[1] Plan 2 building block LevelSetIdentity. Exact deficit identity for finite-perimeter superlevel sets; Lemmas 8.1–8.3 (profile-gap measure, superlevel replacement).

[2] Plan 2 building block MetricFramework. Metric space $\mathcal{X}$, metric first variation, per-ρ velocity defect.

[3] Plan 2 building block CentroidBound. Auxiliary centroid estimates for the alternative space–time route.

[4] Plan 2 building block SpaceTimeIdentity. Space-time Π identity and auxiliary centroid-route diagnostics.

[5] Plan 2 building block WeightedMetricTrace. Integrated action $\to$ pointwise endpoint at $\widehat{\rho}$; metric trace estimate $d_{\mathcal{X}}(F_{\widehat{\rho}}, B_1)^2 \le C_*\delta_T$ with $\widehat{\rho} \in [\rho_\delta - C_*\delta_T, \rho_\delta]$.

[6] Plan 2 building block TrimmedVelocityRepair. Self-truncated velocity defect replacing the pointwise lower-gradient hypothesis.

[7] Plan 2 building block GlobalAssembly. Bounded reduction + Kohler–Jobin transfer; final sharp Faber–Krahn inequality.

[8] `Final/BoundedReduction.tex` (Plan 1 import). Bounded reduction theorem at $R = R_*(n)$.

[9] `Final/KohlerJobinTransfer.tex` (Plan 1 import). Kohler–Jobin transfer from sharp Saint–Venant to sharp Faber–Krahn.

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